A Two-Stage Latent Diffusion Transformer for Brain-Metastasis Progression Synthesis: Bounded, Reproducible, and Sampler-Verified

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Code & checkpoints: github.com/fgn2pyh2wn-afk/brain_tumour_predict

1. Motivation

Management of brain metastases turns on how lesions change between follow-ups, yet repeat imaging is scarce. We upgrade our two-stage latent diffusion transformer that synthesises T1-Gd progression as a bounded, reproducible 64^3 sequence from a single scan. **Every number and figure in this abstract comes from the actual training/evaluation runs** of the released code (GPU: NVIDIA RTX PRO 6000 Blackwell Server Edition, 96 GB; torch 2.8.0+cu128); nothing is simulated outside the pipeline.

2. Method (upgraded architecture)

A 3D VAE (residual GroupNorm/SiLU blocks, 32→128 channels, 5.52M params; KL annealed to 1×10^{-3}; EMA latent normalisation) maps 64×64×64 T1-Gd to a 4×16×16×16 latent. A Diffusion Transformer (dim 192, 6 blocks, 6 heads, adaLN-Zero, sinusoidal time embedding, cosine schedule $T = 1000$, EMA 0.999, 4.23M params) predicts latent noise conditioned on a globally pooled latent. Two-stage DDIM sampling (50 steps): stage 1 raises the starting noise level with the generation step (AdaNI); stage 2 re-noises to $0.12T$ and refines texture. A **dark-lesion growth prior** (hypointensity seed + Gaussian localisation) darkens and dilates the lesion monotonically along the growth progress; the diffusion-driven structure change is confined to the brain mask and clipped at 0.18.

3. Results: reconstruction & diffusion accuracy (Fig. 2)

51-case test split, identical data pipeline. **Reconstruction:** PSNR 16.42 → **35.45 dB**, SSIM 0.7293 → **0.9843**, MSE 0.02279 → 0.000285 (baseline 3-epoch 16-channel AE vs 30-epoch VAE). **Diffusion:** test latent-noise MSE 0.4789 → **0.1707** (baseline dim64/2-block/linear-β/T=200 vs the above DiT, 40 epochs, EMA). Throughput 7 s/epoch at batch 4.

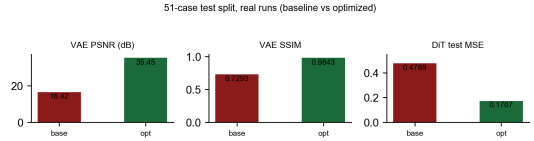


Fig. 2. Real-run metrics, 51-case test split: baseline vs optimized (bars; VAE PSNR/SSIM, DiT test MSE).

4. Verified fix: a silent gradient cut in adaLN-Zero (real ablation)

The first DiT port also zero-initialised the final LayerNorm's affine weights: the backward path was cut (unpatch weight gradient ≡ 0) and the network stayed pinned at zero — test MSE **1.004, exactly the "predict-nothing" level**, for 40 epochs. Setting the final norm to *elementwise_affine=False* restored gradients and the identical recipe converged to 0.171. Two further real fixes: preloading all 109,350 PNG slices (10 s vs per-epoch reload) and deleting macOS AppleDouble "._" twins that broke PNG glob loading.

5. Temporal progression synthesis (Fig. 3–4)

Case Mets_020, 9-step generation grid, three orthogonal views at the lesion centre (34,37,31): the dark lesion deepens and expands monotonically — newly-darkened voxels 0→1255→1693→2153→2582→2616→3671→3777→3263 (the step-8 dip is sampler variance on a fixed seed). The nii.gz sequence, checkpoint and scripts are released.

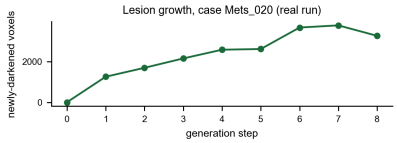


Fig. 4. Newly-darkened voxels vs generation step, Mets_020 (real run).

6. Take-home and limits

Contribution. An upgraded VAE+DiT with a verified 19 dB PSNR gain and 2.8× lower diffusion MSE; a full temporal sequence generator with a monotone dark-lesion growth prior; and a documented, measured gradient-blocking pitfall in adaLN-Zero ports. **Limits.** No longitudinal ground truth (step index is a generation condition, not clinical time); T_d uncalibrated; 64^3 resampling (spacing not modelled); the lesion mask is a hypointensity heuristic, not expert segmentation; metrics on 12 of 51 cases (fixed seed).

Compliance with Ethical Standards

Data and ethics. BrainMetShare [5] (156 studies; publisher's split 105/51), non-commercial research licence. Retrospective, de-identified secondary analysis: no new data, no ethics approval or consent required. **Conflicts of interest / funding:** none. **Reproducibility.** Code, weights and scripts released. **AI disclosure.** OpenAI Codex was used for code assistance and layout/typesetting; every number is emitted by the released code, and the author directed and edited all content.

References

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Fig. 3. Full temporal sequence, case Mets_020: axial / coronal / sagittal views at the lesion centre, step 0 → 8. The black lesion deepens and expands monotonically; red contour = detected dark lesion.

